THE FIRST COHOMOLOGY OF THE INTERNAL HOM OF A LOCALLY PROJECTIVE MODULE VANISHES ON AN AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4802-4812 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4814-4834, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Quasi-coherent modules on the small etale site, locally projective modules, and the internal hom sheaf; the reduction is retained.

Source reference. Original lemma coherent-lemma-quasi-coherent-affine-cohomology-zero: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4834–4834. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma descent-proposition-equivalence-quasi-coherent: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4834–4834. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma descent-proposition-same-cohomology-quasi-coherent: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4834–4834. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma sites-cohomology-lemma-cohomology-products: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4834–4834. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-definition-locally-projective: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4834–4834. Cited by the retained statement or proof but not part of the collected spans;

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

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Lemma 0.1. *Let T be an affine scheme. Let $\mathcal{F}$, $\mathcal{G}$ be quasi-coherent $\mathcal{O}_T$-modules on $T_{étale}$. Consider the internal hom sheaf $\mathcal{H} = \mathcal{H}om_{\mathcal{O}_T}(\mathcal{F}, \mathcal{G})$ on $T_{étale}$. If $\mathcal{F}$ is locally projective, then $H^1(T_{étale}, \mathcal{H}) = 0$.*

Proof. By the definition of a locally projective sheaf on an algebraic space (see Properties of Spaces, Definition spaces-properties-definition-locally-projective) we see that $\mathcal{F}_{Zar} = \mathcal{F}|_{T_{Zar}}$ is a locally projective sheaf on the scheme T. Thus $\mathcal{F}_{Zar}$ is a direct summand of a free $\mathcal{O}_{T_{Zar}}$-module. Whereupon we conclude (as $\mathcal{F} = (\mathcal{F}_{Zar})^a$, see Descent, Proposition descent-proposition-equivalence-quasi-coherent) that $\mathcal{F}$ is a direct summand of a free $\mathcal{O}_T$-module on $T_{étale}$. Hence we may assume that $\mathcal{F} = \bigoplus_{i \in I} \mathcal{O}_T$ is a free module. In this case $\mathcal{H} = \prod_{i \in I} \mathcal{G}$ is a product of quasi-coherent modules. By Cohomology on Sites, Lemma sites-cohomology-lemma-cohomology-products we conclude that $H^1 = 0$ because the cohomology of a quasi-coherent sheaf on an affine scheme is zero, see Descent, Proposition descent-proposition-same-cohomology-quasi-coherent and Cohomology of Schemes, Lemma coherent-lemma-quasi-coherent-affine-cohomology-zero. □